19. Install a C compiler in the virtual machine created using virtual box / VM Ware Workstation / Player and execute Simple C Programs.

AIM:

To install the GCC C compiler in the Ubuntu virtual machine created using Oracle VM VirtualBox and execute simple C programs.

PROCEDURE

- Start the Ubuntu virtual machine in Oracle VM VirtualBox.
- Open the Terminal using Ctrl + Alt + T.
- Update the package list using:

Bash

sudo apt update Install the GCC compiler:

Bash

sudo apt install gcc -y Verify the installation:

Bash

gcc --version Create a C source file:

Bash

nano hello.c Enter a simple C program to display Hello World.

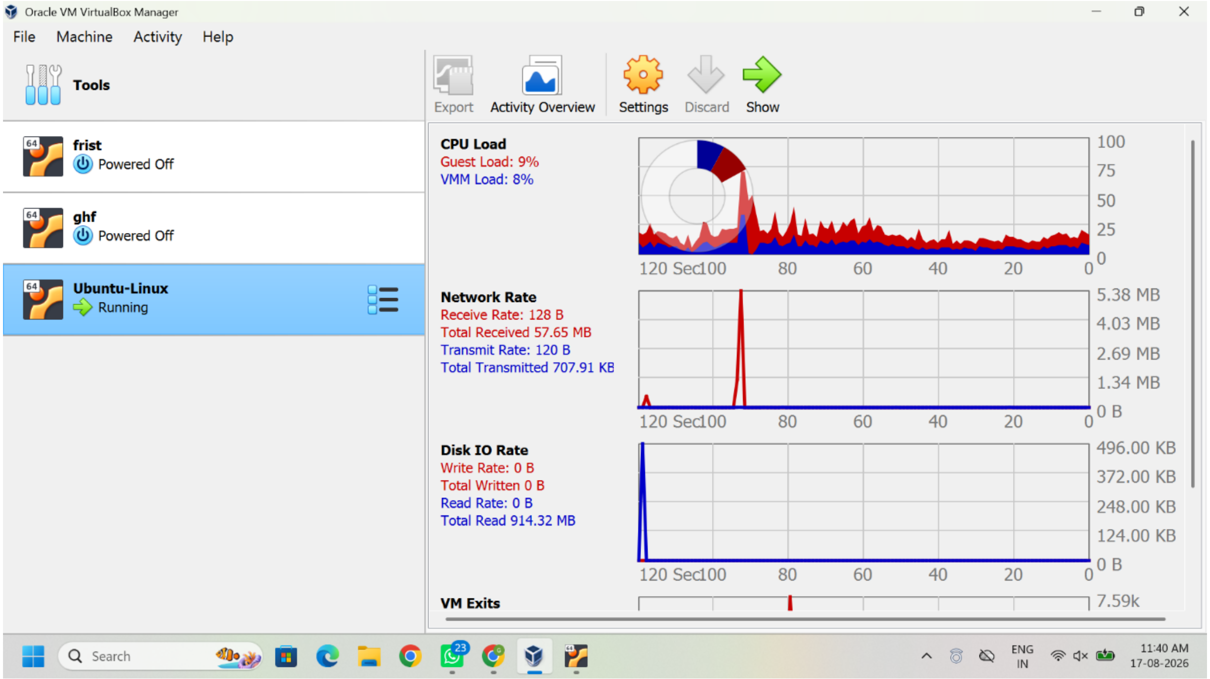
- Save the program using Ctrl + O, press Enter, and exit using Ctrl + X.
- Compile the C program:

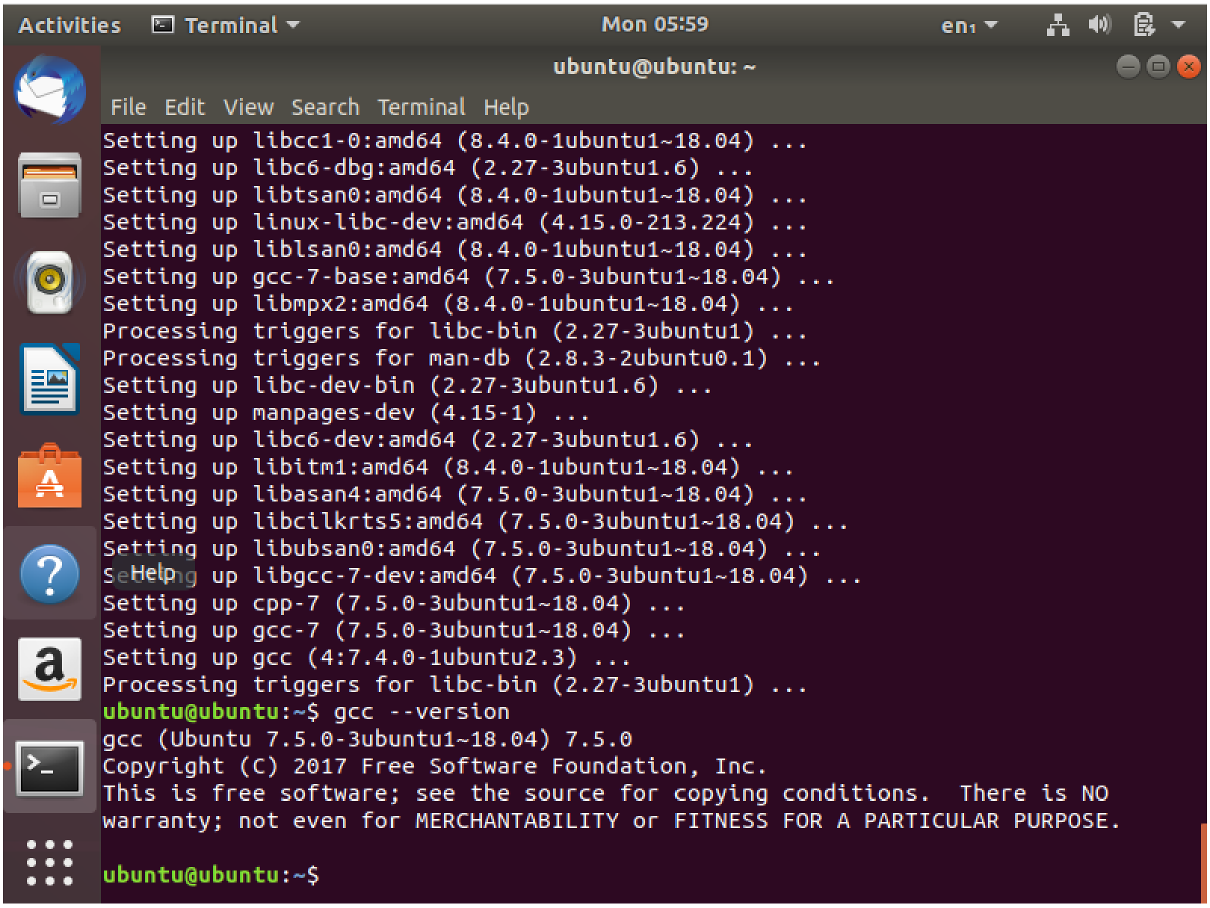
Bash

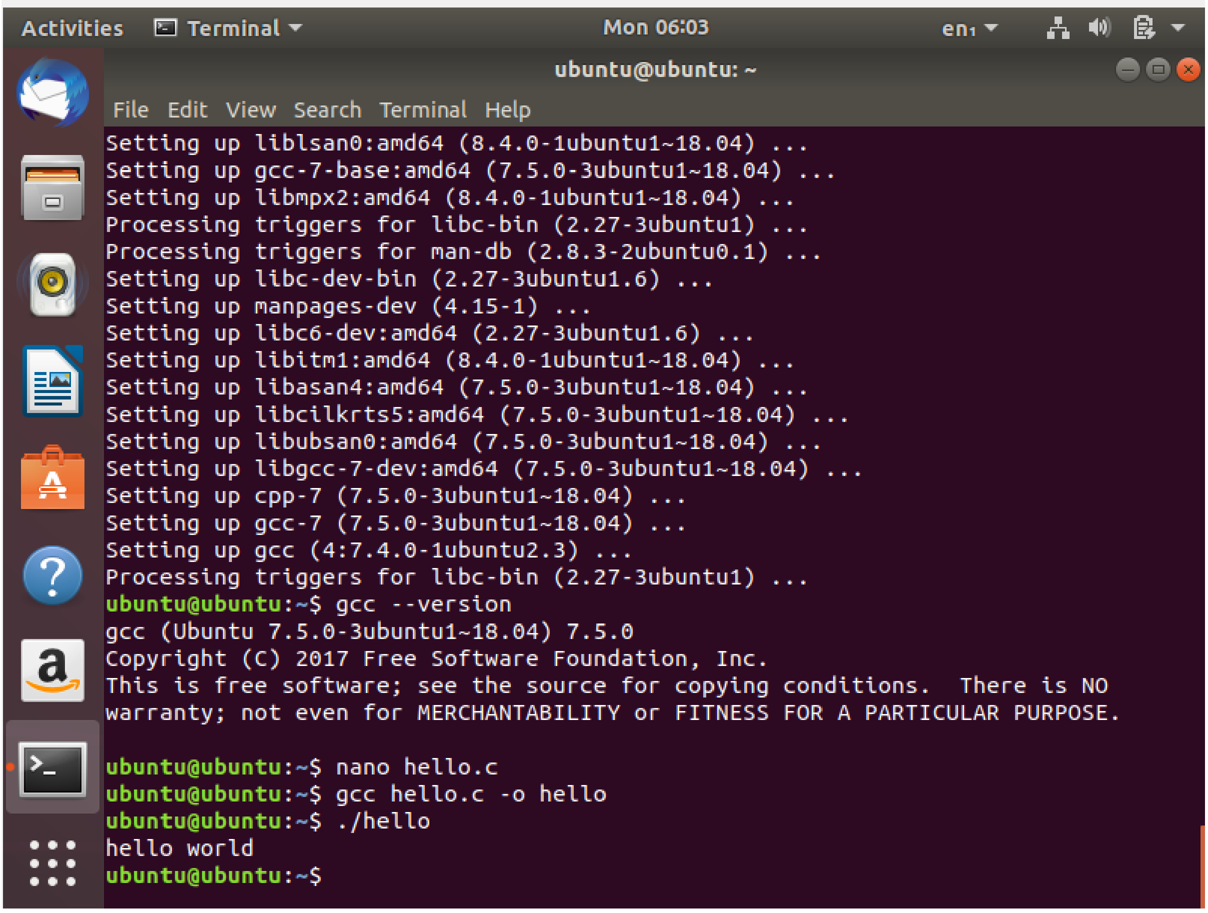
gcc hello.c -o hello Execute the compiled program:

Bash

./hello Verify that the expected output is displayed in the terminal.


Oracle VM VirtualBox Manager
File Machine Activity Help
Tools
Export Activity Overview Settings Discard Show
frist
Powered Off
ghf
Powered Off
Ubuntu-Linux
Running
CPU Load
Guest Load: 9%
VMM Load: 8%
Network Rate
Receive Rate: 128 B
Total Received 57.65 MB
Transmit Rate: 120 B
Total Transmitted 707.91 KB
Disk IO Rate
Write Rate: 0 B
Total Written 0 B
Read Rate: 0 B
Total Read 914.32 MB
VM Exits
100 75 50 25 0
5.38 MB 4.03 MB 2.69 MB 1.34 MB 0 B
496.00 KB 372.00 KB 248.00 KB 124.00 KB 0 B
7.59k
120 Sec 100 80 60 40 20 0
Search
ENG IN
11:40 AM
17-08-2026


Activities Terminal Mon 05:59 en₁
ubuntu@ubuntu: ~
File Edit View Search Terminal Help
Setting up libcc1-0:amd64 (8.4.0-1ubuntu1~18.04) ...
Setting up libc6-dbg:amd64 (2.27-3ubuntu1.6) ...
Setting up libtsan0:amd64 (8.4.0-1ubuntu1~18.04) ...
Setting up linux-libc-dev:amd64 (4.15.0-213.224) ...
Setting up liblsan0:amd64 (8.4.0-1ubuntu1~18.04) ...
Setting up gcc-7-base:amd64 (7.5.0-3ubuntu1~18.04) ...
Setting up libmpx2:amd64 (8.4.0-1ubuntu1~18.04) ...
Processing triggers for libc-bin (2.27-3ubuntu1) ...
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
Setting up libc-dev-bin (2.27-3ubuntu1.6) ...
Setting up manpages-dev (4.15-1) ...
Setting up libc6-dev:amd64 (2.27-3ubuntu1.6) ...
Setting up libitm1:amd64 (8.4.0-1ubuntu1~18.04) ...
Setting up libasan4:amd64 (7.5.0-3ubuntu1~18.04) ...
Setting up libcilkrts5:amd64 (7.5.0-3ubuntu1~18.04) ...
Setting up libubsan0:amd64 (7.5.0-3ubuntu1~18.04) ...
Setting up libgcc-7-dev:amd64 (7.5.0-3ubuntu1~18.04) ...
Setting up cpp-7 (7.5.0-3ubuntu1~18.04) ...
Setting up gcc-7 (7.5.0-3ubuntu1~18.04) ...
Setting up gcc (4:7.4.0-1ubuntu2.3) ...
Processing triggers for libc-bin (2.27-3ubuntu1) ...
ubuntu@ubuntu:~$ gcc --version
gcc (Ubuntu 7.5.0-3ubuntu1~18.04) 7.5.0
Copyright (C) 2017 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
ubuntu@ubuntu:~$
Help



RESULT

The GCC C compiler was successfully installed in the Ubuntu virtual machine, and the simple C program was successfully compiled and executed.